

IIT HYDERABAD

DEPARTMENT OF ELECTRICAL ENGINEERING

Digital Systems Laboratory Report

Assignment No: **03**



Course Name: Digital Systems Lab

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1 Objective

The objective of this experiment is to design and implement a 4-bit synchronous up-down counter using **Verilog** on the Real Digital Boolean Board (FPGA Board). The implementation focuses on utilizing a mode switch to control counting direction, integrating a manual reset button for state initialization, and mapping the 4-bit output to on-board LEDs to visually verify real-time state transitions synchronized with the system clock.

2 Design Methodology

The implementation follows a modular behavioral approach in Verilog, consisting of two main functional stages :

2.1 Clock design

The Boolean Board provides a high-speed 100MHz system clock. To make the state transitions visible on the LEDs, a 27-bit register (`clk_div`) is used as a counter.

- It counts up to 50,000,000 cycles. (But , we ran the simulation for only 4 cycles inorder to observe clear and noticeable transition)
- Upon reaching this limit, it toggles the `slow_clk` signal and resets.
- This results in a `slow_clk` frequency of exactly 1 Hz.

2.2 Sequential Counter design

The counter operates on the positive edge of the `slow_clk`.

- The counter is programmed such that `mode = 0` triggers an increment (Up) and `mode = 1` triggers a decrement (Down).
- If reset is triggered while in Up-mode, it returns to 0000. If triggered in Down-mode, it pre-loads 1111 to allow for a full downward cycle.

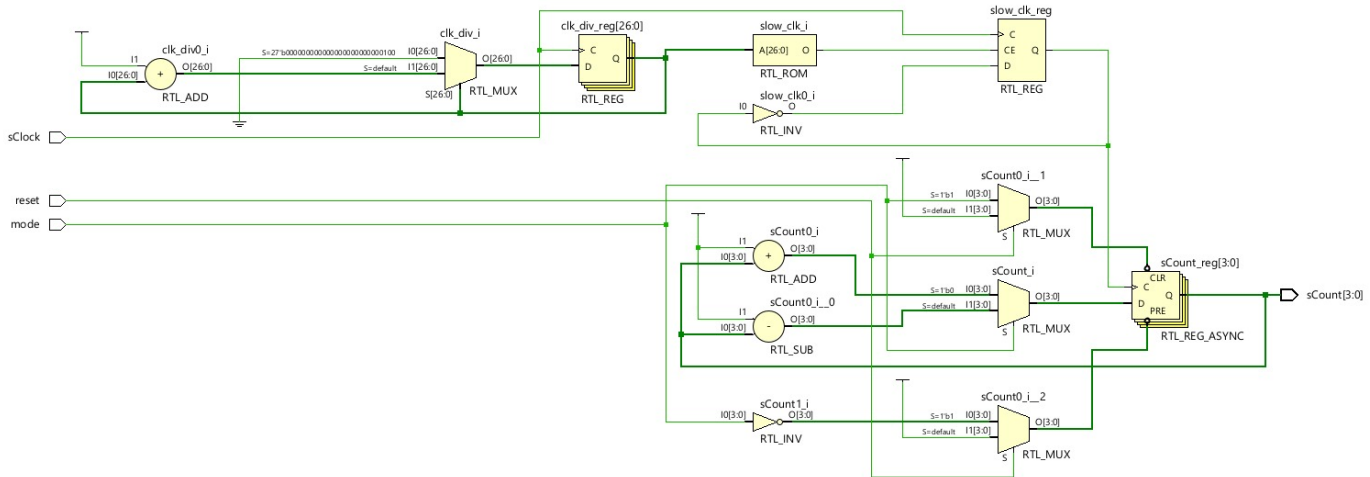


Figure 1: Schematic

3 Code Implementation

Design code :

```

module UpDown_Counter(
    input sClock ,
    input reset ,
    input mode,
    output reg [3:0] sCount = 4'b0000
);

    reg [26:0] clk_div = 0;
    reg slow_clk = 0;

    always @(posedge sClock) begin
        if (clk_div == 50_000_000) begin
            clk_div <= 0;
            slow_clk <= ~slow_clk;
        end else begin
            clk_div <= clk_div + 1;
        end
    end

    always @(posedge slow_clk or posedge reset) begin

```

```
        if (reset) begin
            if (mode==1'b0)begin
                sCount <= 4'b0000;
            end
            else begin
                sCount <= 4'b1111;
            end
        end else begin
            if (mode==1'b0) begin
                sCount <= sCount + 1;
            end else begin
                sCount <= sCount - 1;
            end
        end
    end
end
endmodule
```

Testbench code :

```
module UpDown_Counter_tb();
reg sClk;
reg rst;
reg mod;
wire [3:0] cont;

UpDown_Counter uut(.sClock(sClk), .reset(rst), .mode(mod), .sCount(cont));

initial sClk = 0;
    always #5 sClk = ~sClk;

initial begin
    rst = 0;
    mod = 0;
    #300;
    rst = 1;
    #1;
    rst = 0;
    #500;
    mod = 1;
end
```

```

    rst = 1;
    #1;
    rst = 0;
    #500;
    $finish;
end
endmodule

```

4 Simulation and Verification

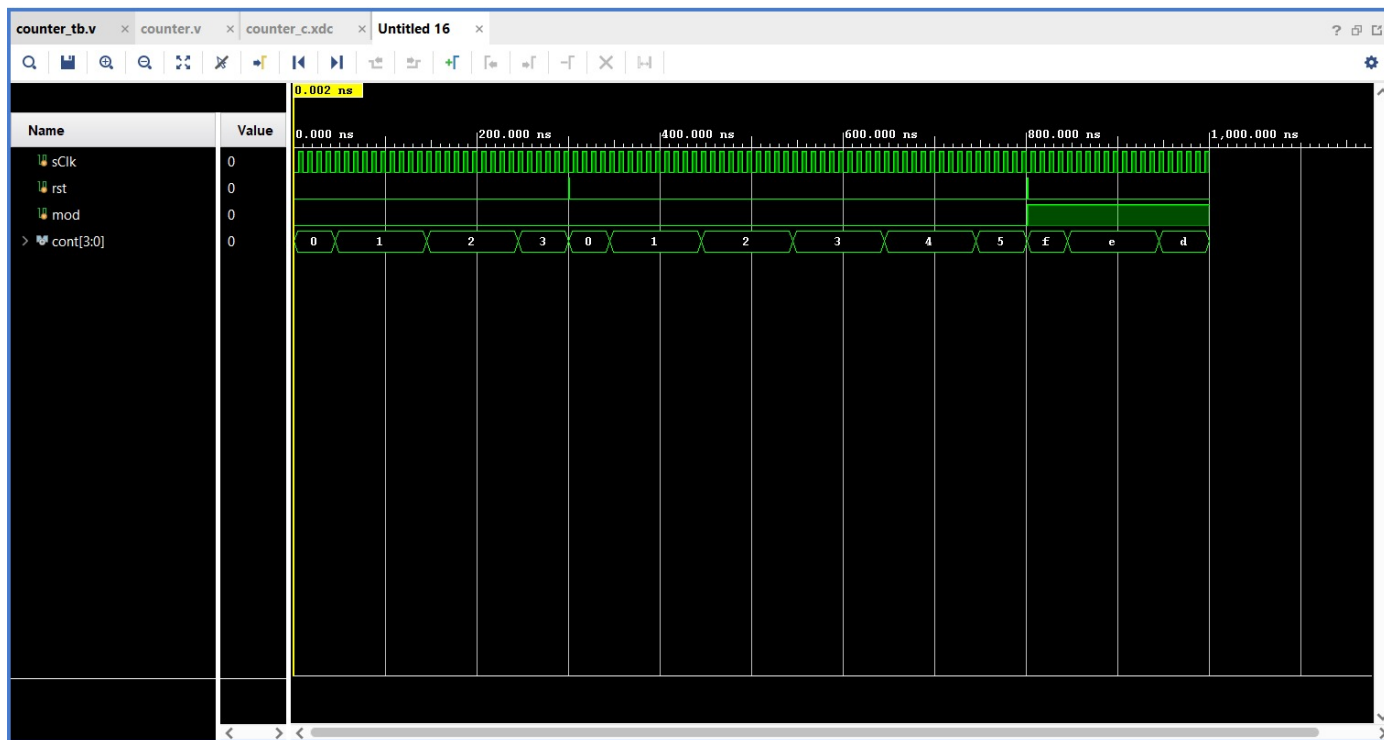


Figure 2: Simulation

The functional verification was performed using a Verilog testbench, as shown in the waveform. Initially, with the reset (*rst*) and mode (*mod*) signals at logic low, the counter (*cont[3:0]*) correctly increments from 0 through 3 on consecutive clock edges. At approximately 400ns, the *rst* signal is asserted, which immediately clears the counter to 0, verifying the high-priority asynchronous reset logic. Upon de-assertion of the reset, the counter resumes its upward sequence ($0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5$). At the 800ns mark, the *mod* signal transitions to logic high; the subsequent clock edges demonstrate the counter shifting into its decrementing state ($F \rightarrow E \rightarrow D$), confirming that the directional control logic and the "mode-aware" reset transitions are functioning as intended.

5 Hardware Implementation

Constraints :

```
set_property PACKAGE_PIN A4 [get_ports {sCount[3]}]
set_property PACKAGE_PIN B4 [get_ports {sCount[2]}]
set_property PACKAGE_PIN A3 [get_ports {sCount[1]}]
set_property PACKAGE_PIN B3 [get_ports {sCount[0]}]
set_property PACKAGE_PIN K1 [get_ports mode]
set_property PACKAGE_PIN K2 [get_ports reset]
set_property PACKAGE_PIN F14 [get_ports sClock]
set_property IOSTANDARD LVCMOS33 [get_ports {sCount[3]}]
set_property IOSTANDARD LVCMOS33 [get_ports {sCount[2]}]
set_property IOSTANDARD LVCMOS33 [get_ports {sCount[1]}]
set_property IOSTANDARD LVCMOS33 [get_ports {sCount[0]}]
set_property IOSTANDARD LVCMOS33 [get_ports mode]
set_property IOSTANDARD LVCMOS33 [get_ports reset]
set_property IOSTANDARD LVCMOS33 [get_ports sClock]
```

The design was mapped to the Spartan-7 FPGA on the Boolean Board using the following physical pin constraints :

Signal Name	FPGA Pin	I/O Standard	Hardware Component
sClock	F14	LVCMOS33	100MHz System Clock
reset	K2	LVCMOS33	Slide Switch (Reset)
mode	K1	LVCMOS33	Slide Switch (Direction)
sCount[3]	A4	LVCMOS33	Output Bit 3 (MSB)
sCount[2]	B4	LVCMOS33	Output Bit 2
sCount[1]	A3	LVCMOS33	Output Bit 1
sCount[0]	B3	LVCMOS33	Output Bit 0 (LSB)

6 Results and Observations

The working demonstration of the circuit is available at [DEMO](#).

- The clock divider circuit generated a slower clock signal from the high-frequency system clock, making the counter transitions visibly observable.
- When the `mode = 0`, the counter operated in **up-counting mode**, and the LED outputs incremented sequentially from 0000 to 1111.

- When the `mode = 1`, the counter operated in **down-counting mode**, and the LED outputs decremented sequentially from 1111 to 0000.
- On pressing the reset button:
 - For `mode = 0`, the counter reset to 0000.
 - For `mode = 1`, the counter reset to 1111.
- The transitions between states were smooth and synchronized with the slow clock signal.
- No glitches or unexpected outputs were observed during operation.

7 Conclusion

The Up-Down Counter was successfully designed and implemented using behavioral modeling in Verilog. The counter operated correctly in both incrementing and decrementing modes based on the control input. The clock divider enabled clear observation of state transitions on the FPGA LEDs. This experiment demonstrated the practical working of sequential logic and clock-driven circuits in an FPGA environment.